

Matrix Groups

Road map

Matrix groups provide the central examples for Lie theory. We define the general and special linear groups, the orthogonal and unitary groups and their special subgroups, and briefly recall the symplectic group. Determinant homomorphisms identify the basic quotient groups. We then analyze scalar multiplication maps in detail, obtaining decompositions of $GL(n, \mathbb{R})$, $U(n)$, and $GL(n, \mathbb{C})$, with separate conclusions in odd and even real dimensions.

10.1 Fields, matrices, and characteristic

Throughout this chapter,

$$K = \mathbb{R} \quad \text{or} \quad K = \mathbb{C}.$$

Both fields have characteristic zero. In general, the *characteristic* of a field K , denoted $\text{char } K$, is the least positive integer m for which

$$\underbrace{1_K + \cdots + 1_K}_{m \text{ terms}} = 0_K,$$

if such an integer exists; otherwise the characteristic is 0. Thus no positive multiple of 1 vanishes in $\mathbb{R}$ or $\mathbb{C}$.

Let

$$\text{Mat}(n, K)$$

be the set of all $n \times n$ matrices with entries in K . Under matrix multiplication this set is not a group, because a singular matrix has no multiplicative inverse.

10.2 The general and special linear groups

Definition 10.1: General linear group

$$GL(n, K) = \{A \in \text{Mat}(n, K) : \det A \neq 0\}.$$

It is called the *general linear group of degree n over K* .

Proposition 10.1

$GL(n, K)$ is a group under matrix multiplication.

证明. Associativity is inherited from matrix multiplication. If $A, B \in GL(n, K)$, then

$$\det(AB) = \det A \det B \neq 0,$$

so $AB \in GL(n, K)$. The identity matrix has determinant 1, and therefore lies in the group. Finally,

$$\det A \neq 0 \iff A \text{ is invertible,}$$

and

$$\det(A^{-1}) = (\det A)^{-1} \neq 0.$$

Thus $A^{-1} \in GL(n, K)$. ■

Definition 10.2: Special linear group

$$SL(n, K) = \{A \in GL(n, K) : \det A = 1\}.$$

It is called the *special linear group of degree n over K* .

The word “special” records the additional determinant-one condition. The subgroup inclusions are

$$\begin{array}{ccc} SL(n, \mathbb{R}) & < & GL(n, \mathbb{R}), \\ \cap & & \cap \\ SL(n, \mathbb{C}) & < & GL(n, \mathbb{C}), \end{array}$$

where the vertical inclusions regard a real matrix as a complex matrix.

10.2.1 The determinant homomorphism

Let

$$K^\times = K \setminus \{0\}$$

be the multiplicative group of nonzero field elements.

Theorem 10.1

$$SL(n, K) \trianglelefteq GL(n, K)$$

and

$$GL(n, K)/SL(n, K) \cong K^\times.$$

证明. Consider

$$\det : GL(n, K) \longrightarrow K^\times.$$

The target is a group under multiplication, with identity 1 and inverse $\lambda^{-1} = 1/\lambda$. The determinant is a homomorphism:

$$\det(AB) = \det A \det B.$$

Its kernel is exactly

$$\ker(\det) = \{A \in GL(n, K) : \det A = 1\} = SL(n, K).$$

Therefore $SL(n, K)$ is normal.

The determinant map is surjective. Given $\lambda \in K^\times$, the diagonal matrix

$$D_\lambda = \text{diag}(1, \dots, 1, \lambda)$$

is invertible and satisfies $\det D_\lambda = \lambda$. Hence the First Isomorphism Theorem gives

$$GL(n, K)/SL(n, K) \cong K^\times.$$

■

10.2.2 The quotient isomorphism written directly

The induced map is

$$\tau : GL(n, K)/SL(n, K) \longrightarrow K^\times, \quad \tau(A SL(n, K)) = \det A.$$

It is a homomorphism because

$$\begin{aligned} \tau((A SL)(B SL)) &= \tau(AB SL) \\ &= \det(AB) \\ &= \det A \det B \\ &= \tau(A SL)\tau(B SL). \end{aligned}$$

It is well defined and injective because

$$\begin{aligned} \tau(A SL) = \tau(B SL) &\iff \det A = \det B \\ &\iff \det(A^{-1}B) = 1 \\ &\iff A^{-1}B \in SL(n, K) \\ &\iff A SL(n, K) = B SL(n, K). \end{aligned}$$

Finally, the matrices D_λ above show directly that τ is surjective.

10.3 Orthogonal, unitary, and symplectic groups

For a complex matrix A , write

$$A^\dagger = \overline{A}^T$$

for its conjugate transpose.

Definition 10.3: Orthogonal group

$$O(n) = \{A \in GL(n, \mathbb{R}) : AA^T = I_n\}.$$

The equation $AA^T = I$ is equivalent to $A^{-1} = A^T$, and also to $A^T A = I$. If $A, B \in O(n)$, then

$$(AB)(AB)^T = ABB^T A^T = AA^T = I,$$

and $A^{-1} = A^T \in O(n)$. Thus $O(n)$ is a group.

Definition 10.4: Unitary group

$$U(n) = \{A \in GL(n, \mathbb{C}) : AA^\dagger = I_n\}.$$

The same calculation, with T replaced by $\dagger$, proves that $U(n)$ is a group. A unitary matrix satisfies

$$A^{-1} = A^\dagger.$$

Definition 10.5: Symplectic group

Let Γ be a fixed real nondegenerate skew-symmetric $2n \times 2n$ matrix. Then

$$\mathrm{Sp}(2n, \mathbb{R}) = \{A \in GL(2n, \mathbb{R}) : A^T \Gamma A = \Gamma\}.$$

As proved in Chapter 8, this is a group, its isomorphism type is independent of the chosen nondegenerate skew form, and

$$\det A = 1 \quad (A \in \mathrm{Sp}(2n, \mathbb{R})).$$

The symplectic group is the structure-preserving linear group of classical Hamiltonian dynamics.

10.4 Special orthogonal and special unitary groups

10.4.1 The orthogonal determinant

If $A \in O(n)$, then

$$1 = \det(AA^T) = \det A \det A^T = (\det A)^2.$$

Thus $\det A = \pm 1$, and the determinant restricts to a homomorphism

$$\det : O(n) \longrightarrow \{+1, -1\} \cong \mathbb{Z}_2.$$

Definition 10.6: Special orthogonal group

$$SO(n) = \{A \in O(n) : \det A = 1\}.$$

The kernel of the restricted determinant is $SO(n)$. The map is surjective because

$$\mathrm{diag}(-1, 1, \dots, 1) \in O(n)$$

has determinant -1 . Therefore

$$SO(n) \trianglelefteq O(n), \quad \boxed{O(n)/SO(n) \cong \mathbb{Z}_2.}$$

10.4.2 The unitary determinant

If $A \in U(n)$, then

$$1 = \det(AA^\dagger) = \det A \det A^\dagger = \det A \overline{\det A} = |\det A|^2.$$

Hence $\det A$ lies on the unit circle

$$U(1) = \{e^{i\theta} : 0 \leq \theta < 2\pi\}.$$

Definition 10.7: Special unitary group

$$SU(n) = \{A \in U(n) : \det A = 1\}.$$

The determinant homomorphism

$$\det : U(n) \longrightarrow U(1)$$

is surjective: for every $e^{i\theta} \in U(1)$, the diagonal matrix

$$\text{diag}(1, \dots, 1, e^{i\theta})$$

is unitary and has that determinant. Its kernel is $SU(n)$. Thus

$$SU(n) \trianglelefteq U(n), \quad \boxed{U(n)/SU(n) \cong U(1)}.$$

10.5 Scalar matrices and real general linear groups

Let

$$\overline{\mathbb{R}^\times} = \{\lambda I_n : \lambda \in \mathbb{R}^\times\} \leq GL(n, \mathbb{R}).$$

This scalar subgroup is isomorphic to $\mathbb{R}^\times$ and is central, hence normal.

10.5.1 Odd dimension

Theorem 10.2: Odd-dimensional real decomposition

If n is odd, then

$$\boxed{GL(n, \mathbb{R}) \cong SL(n, \mathbb{R}) \times \mathbb{R}^\times}.$$

More precisely, the multiplication map

$$\kappa : SL(n, \mathbb{R}) \times \mathbb{R}^\times \longrightarrow GL(n, \mathbb{R}), \quad \kappa(A, \lambda) = \lambda A,$$

is an isomorphism.

证明. Scalar matrices commute with every matrix, so

$$\kappa((A, \lambda)(B, \mu)) = \lambda\mu AB = (\lambda A)(\mu B) = \kappa(A, \lambda)\kappa(B, \mu).$$

Thus κ is a homomorphism.

If $\kappa(A, \lambda) = I$, then $A = \lambda^{-1}I$. Since $A \in SL(n, \mathbb{R})$,

$$1 = \det A = \det(\lambda^{-1}I) = \lambda^{-n}.$$

For odd n , the real equation $\lambda^n = 1$ has the unique solution $\lambda = 1$. Hence $A = I$, and the kernel is trivial.

Finally, let $B \in GL(n, \mathbb{R})$. Because n is odd, the real number

$$\lambda = (\det B)^{1/n}$$

exists and is nonzero even when $\det B < 0$. Set

$$A = \lambda^{-1}B.$$

Then

$$\det A = \lambda^{-n} \det B = 1,$$

so $A \in SL(n, \mathbb{R})$, and $\kappa(A, \lambda) = B$. Thus κ is surjective and therefore an isomorphism. ■

This is also an internal direct product:

$$SL(n, \mathbb{R}) \cap \overline{\mathbb{R}^\times} = \{I\}, \quad SL(n, \mathbb{R}) \overline{\mathbb{R}^\times} = GL(n, \mathbb{R})$$

when n is odd.

10.5.2 Even dimension

For even n , the same map is not onto all of $GL(n, \mathbb{R})$, because

$$\det(\lambda A) = \lambda^n \det A = \lambda^n > 0$$

when $A \in SL(n, \mathbb{R})$. Define

$$GL_+(n, \mathbb{R}) = \{B \in GL(n, \mathbb{R}) : \det B > 0\}.$$

Theorem 10.3: Even-dimensional positive-determinant decomposition

If n is even, then

$$\frac{SL(n, \mathbb{R}) \times \mathbb{R}^\times}{\mathbb{Z}_2} \cong GL_+(n, \mathbb{R}),$$

where the embedded $\mathbb{Z}_2$ is

$$\{(I, 1), (-I, -1)\}.$$

证明. Use the homomorphism

$$\kappa(A, \lambda) = \lambda A.$$

If $\kappa(A, \lambda) = I$, then $A = \lambda^{-1}I$ and

$$1 = \det A = \lambda^{-n}.$$

For real $\lambda \neq 0$ and even n , the equation $\lambda^n = 1$ has the two solutions $\lambda = \pm 1$. Thus

$$\ker \kappa = \{(I, 1), (-I, -1)\} \cong \mathbb{Z}_2.$$

We already observed that $\text{im } \kappa \subset GL_+(n, \mathbb{R})$. Conversely, let $B \in GL_+(n, \mathbb{R})$ and put

$$d = \det B > 0, \quad \lambda = d^{1/n} > 0, \quad A = \lambda^{-1}B.$$

Then $\det A = d^{-1} \det B = 1$, and $\kappa(A, \lambda) = B$. Therefore $\text{im } \kappa = GL_+(n, \mathbb{R})$. The First Isomorphism Theorem now gives the stated quotient. ■

Why this differs from $GL(n, \mathbb{R})/SL(n, \mathbb{R})$

For every n ,

$$GL(n, \mathbb{R})/SL(n, \mathbb{R}) \cong \mathbb{R}^\times$$

via the determinant. The scalar multiplication map is a different homomorphism. In even dimension its image misses all negative-determinant matrices and its kernel contains the simultaneous sign change $(A, \lambda) \mapsto (-A, -\lambda)$.

10.6 The relation between $U(n)$ and $SU(n) \times U(1)$

Define

$$\kappa : SU(n) \times U(1) \longrightarrow U(n), \quad \kappa(S, z) = zS,$$

where z acts as the scalar matrix zI_n .

Proposition 10.2

The map κ is a surjective homomorphism and

$$\ker \kappa = \left\{ \left(e^{-2\pi i k/n} I_n, e^{2\pi i k/n} \right) : k = 0, \dots, n-1 \right\} \cong \mathbb{Z}_n.$$

Consequently,

$$\frac{SU(n) \times U(1)}{\mathbb{Z}_n} \cong U(n).$$

证明. Because a scalar matrix commutes with every S ,

$$\begin{aligned} \kappa((S_1, z_1)(S_2, z_2)) &= \kappa(S_1 S_2, z_1 z_2) \\ &= z_1 z_2 S_1 S_2 \\ &= (z_1 S_1)(z_2 S_2), \end{aligned}$$

so κ is a homomorphism.

If $(S, z) \in \ker \kappa$, then $zS = I$, so

$$S = z^{-1}I.$$

The condition $\det S = 1$ becomes

$$1 = \det(z^{-1}I) = z^{-n},$$

or $z^n = 1$. The n solutions in $U(1)$ are

$$z = e^{2\pi i k/n}, \quad k = 0, \dots, n-1.$$

This gives the displayed kernel. Under multiplication, adding the indices modulo n multiplies the corresponding pairs, so the kernel is isomorphic to the additive cyclic group $\mathbb{Z}_n$.

To prove surjectivity, take $U \in U(n)$. Since $|\det U| = 1$, write

$$\det U = e^{i\theta} \quad (0 \leq \theta < 2\pi).$$

Set

$$z = e^{i\theta/n}, \quad S = z^{-1}U.$$

Then S is unitary, because multiplication by a unit-modulus scalar preserves unitarity, and

$$\det S = z^{-n} \det U = e^{-i\theta} e^{i\theta} = 1.$$

Thus $S \in SU(n)$ and $\kappa(S, z) = U$. The First Isomorphism Theorem finishes the proof. ■

Two related but different quotient statements

$$U(n)/SU(n) \cong U(1)$$

comes from the determinant $U(n) \rightarrow U(1)$, whereas

$$(SU(n) \times U(1))/\mathbb{Z}_n \cong U(n)$$

comes from multiplying a special unitary matrix by a scalar phase. The two statements describe different homomorphisms.

10.7 The complex general linear group

The same construction works with all nonzero complex scalars:

$$\kappa : SL(n, \mathbb{C}) \times \mathbb{C}^\times \longrightarrow GL(n, \mathbb{C}), \quad \kappa(A, z) = zA.$$

Theorem 10.4: Scalar decomposition over $\mathbb{C}$

The map κ is surjective and has kernel

$$\left\{ \left(e^{-2\pi i k/n} I_n, e^{2\pi i k/n} \right) : k = 0, \dots, n-1 \right\} \cong \mathbb{Z}_n.$$

Therefore

$$\frac{SL(n, \mathbb{C}) \times \mathbb{C}^\times}{\mathbb{Z}_n} \cong GL(n, \mathbb{C}).$$

证明. The scalar matrices are central, so κ is a homomorphism. If $zA = I$, then $A = z^{-1}I$, and

$$1 = \det A = z^{-n}.$$

Thus $z^n = 1$, giving exactly the displayed kernel.

For surjectivity, take $B \in GL(n, \mathbb{C})$. Write its nonzero determinant in polar form:

$$\det B = r e^{i\theta}, \quad r > 0, \quad 0 \leq \theta < 2\pi.$$

Choose one n -th root

$$z = r^{1/n} e^{i\theta/n}$$

and set $A = z^{-1}B$. Then

$$\det A = z^{-n} \det B = 1,$$

so $A \in SL(n, \mathbb{C})$, and $\kappa(A, z) = B$. The result follows from the First Isomorphism Theorem. ■

For comparison, the determinant gives the simpler quotient

$$GL(n, \mathbb{C})/SL(n, \mathbb{C}) \cong \mathbb{C}^\times.$$

Again, this is not the same quotient construction as the scalar multiplication theorem.

What to retain from this chapter

1. $GL(n, K)$ consists of the invertible $n \times n$ matrices, and $SL(n, K)$ is its determinant-one normal subgroup.
2. The determinant gives $GL(n, K)/SL(n, K) \cong K^\times$.
3. $O(n)$ and $U(n)$ preserve the real and complex standard inner products; $SO(n)$ and $SU(n)$ impose determinant 1.
4. $O(n)/SO(n) \cong \mathbb{Z}_2$ and $U(n)/SU(n) \cong U(1)$.
5. For odd n , $GL(n, \mathbb{R}) \cong SL(n, \mathbb{R}) \times \mathbb{R}^\times$. For even n , scalar multiplication instead gives

$$(SL(n, \mathbb{R}) \times \mathbb{R}^\times)/\mathbb{Z}_2 \cong GL_+(n, \mathbb{R}).$$

6. Scalar multiplication also gives

$$(SU(n) \times U(1))/\mathbb{Z}_n \cong U(n), \quad (SL(n, \mathbb{C}) \times \mathbb{C}^\times)/\mathbb{Z}_n \cong GL(n, \mathbb{C}).$$

7. The finite kernels $\mathbb{Z}_n$ arise from the n -th roots of unity and record the nonuniqueness of extracting a scalar n -th root of the determinant.